Fig. 2: Electronics labelling has been given a new lease of life with high-resolution micro-printing using high-resolution CIJ printers

is highly difficult to replicate this character formulation.

Dynamically-altered font. Varied numbers and letters within a code can have minute sections of characters missing in order to create distinct codes. The advantage of this code is that, visually it is difficult to recognise the small differences—only supply chain partners who have a trained understanding of the code formation can observe the differences.

Verifiable codes. Tracked with vision-inspection systems, verifiable codes utilise unique software-driven algorithms, which cannot be duplicated without an understanding of the algorithm and keys.

Covert codes. Ultraviolet and infrared inks can be utilised to create covert codes that are not visible to the naked eye. Codes are only visible under ultraviolet or other high-frequency lighting, and these are inconspicuous methods of tracking products throughout the supply chain.

Fig. 3: 1650 and 1620 high-resolution CIJ printers use a 40-micron nozzle to print 2D bar codes and alphanumeric multi-line codes at high speeds on limited areas

Challenges of coding electrical components

Increased productivity and maximised profitability are the main objectives of any production facility, and manufacturers ultimately need to ensure their production line is streamlined in order to safeguard these objectives.

In the electrical manufacturing environment, there are specific challenges related to coding components that the manufacturers must ensure do not interfere with production efficiency. Two of the clearest are limited surface space for marking and coding on small electrical components, and increased complexity of codes.

Given the dynamic nature of modern supply chains, a greater degree of supply chain parties are now demanding that their own specific data is included. This, coupled with anti-counterfeiting coding requirements, means that code complexity has amplified at the same time as components have decreased in size.

Ink durability and compliance are also challenges manufacturers face during the production process. Codes must endure the exacting alcohol cleaning process that electrical components undergo in order to remove solder residue, and need to be durable enough to withstand this process without causing any damage to the components. Further, manufacturers must also ensure that they use halogen-free inks—in line with industry regulations such as Restriction of Hazardous Substance Directive (RoHS).

CIJ, laser printing could provide the solution

Continuous inkjet (CIJ) and laser technologies are perfectly suited to the production and industry requirements of electronic component manufacturing. Both are highly versatile at high speeds and, with smart coding capabilities, CIJ and laser printers' algorithmic software technologies enable manufacturers to create dynamic codes that are difficult to replicate. Also, both printer technologies have the capability to produce highly-legible complex codes on limited surface spaces.

CIJ printers produce high-resolution codes that are extremely durable, even during exacting alcohol cleaning processes. The technology's compatibility with alcohol-resistant, quick-drying and halogen-free inks helps to advance coding and marking applications within the electronic component manufacturing environment, and helps manufacturers to comply with legislative requirements. CIJ printers also offer pigmented inks that offer strong code contrast against electronics with dark surfaces.

Laser printers utilise